



**Faculty of Computer Applications & Information Technology
Integrated MSc(IT) Programme**

“Customer Shopping Behaviour Analysis”

Project Report
submitted for
the partial fulfillment of
Semester – VIII

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Under the Guidance of

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Faculty of Computer Applications & Information Technology
Integrated MSc(IT) Programme

CERTIFICATE

This is to certify that
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students of Semester – VIII Integrated MSc (IT) Programme,
FCAIT, GLS University have successfully completed the
Capstone Project – I
on
“Data Analytics”

as a partial fulfillment of the study of Semester – VIII,
Integrated Master of Science (Information Technology)

Date of Submission: _____

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Introduction:

This project focuses on analyzing a customer purchase dataset that includes information such as age, gender, product category, purchase amount, and shopping behavior. The main objective of this project is to explore the data, identify key patterns, and generate useful insights that can help in improving business strategies and customer satisfaction.

Various data analytics techniques such as data cleaning, preprocessing, exploratory data analysis, and visualization have been applied to understand the dataset effectively. The results of this analysis provide valuable information about customer preferences, purchasing trends, and overall market behavior.

Methodologies:

- **Data Collection:** Dataset was obtained from a reliable source.
- **Data Cleaning:** Missing values and duplicates were handled to improve data quality.
- **Data Preprocessing:** Data was organized and formatted for analysis.
- **Feature Engineering:** New features like age group and total spend were created.
- **Exploratory Data Analysis:** Patterns and trends were identified using statistical methods
- **Data Visualization:** Charts and graphs were used to represent the data clearly.
- **Insight Generation:** Key findings about customer behavior and purchasing trends were derived.

Project Deliverables:

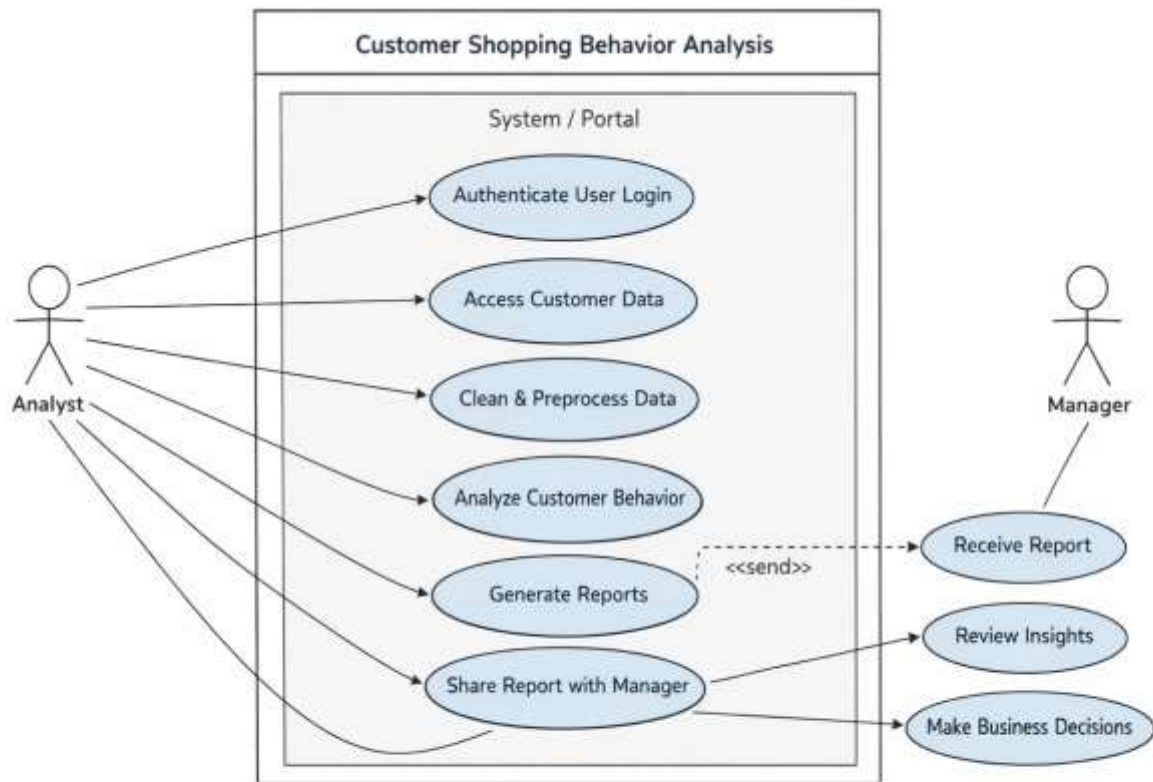
- **Data Preparation & Modeling (Python):** Clean and transform the raw dataset for analysis.
- **Data Analysis (SQL):** Organize the data into a structured format, simulate business transactions, and run queries to extract insights on customer segments, loyalty, and purchase drivers.
- **Visualization & Insights (Power BI):** Build an interactive dashboard that highlights key patterns and trends, enabling stakeholders to make data-driven decisions.
- **Report and Presentation:** Write a clear project report summarizing your key findings and business recommendations. Prepare a presentation that visually communicates insights and actionable recommendations to stakeholders.
- **Streamlit:** An open-source Python framework used to convert data analysis scripts into interactive web applications without requiring any web development knowledge. In this project. It serves as the frontend layer of the web portal rendering interactive charts, KPI metrics, SQL-based business insights, sidebar filters, file upload functionality, and data download options all directly from Python code connected to a MySQL database.

Data Dictionary:

Column Name	Data Type	Description	Example	Remarks
customer_id	Integer	Unique identifier assigned to each customer	1	Primary Key
age	Integer	Age of the customer	25	
gender	String	Gender of the customer	Male	Measured in years
item_purchased	String	Product purchased by the customer	Shirt	Values: Male, Female
category	String	Category of the purchased item	Clothing	e.g. Clothing, Electronics
purchase_amount	Integer	Amount spent on a purchase	500	In INR (₹)
location	String	City of the customer	Mumbai	-
size	String	Size of the purchased item	M	Values: S, M, L, XL
color	String	Color of the purchased product	Blue	-
season	String	Season during which purchase was made	Winter	e.g. Summer, Winter
review_rating	Float	Rating given by the customer	4.5	Scale: 1 to 5
subscription_status	String	Indicates if customer has subscription	Yes	Values: Yes, No
shipping_type	String	Type of shipping selected	Express	Standard / Express
discount_applied	String	Whether discount was applied	Yes	Yes / No
previous_purchases	Integer	Number of past purchases made by customer	10	-
payment_method	String	Payment method used by customer	Credit Card	UPI, Cash, Card
frequency_of_purchases	String	Frequency of customer purchases	Weekly	Weekly, Monthly
age_group	String	Categorized age group of customer	Young Adult	Derived from age
purchase_frequency_days	Float	Average gap (in days) between purchases	14	Derived feature
total_spend	Float	Total amount spent by customer	5000	Aggregated value

UML Diagram:

- Use Case:

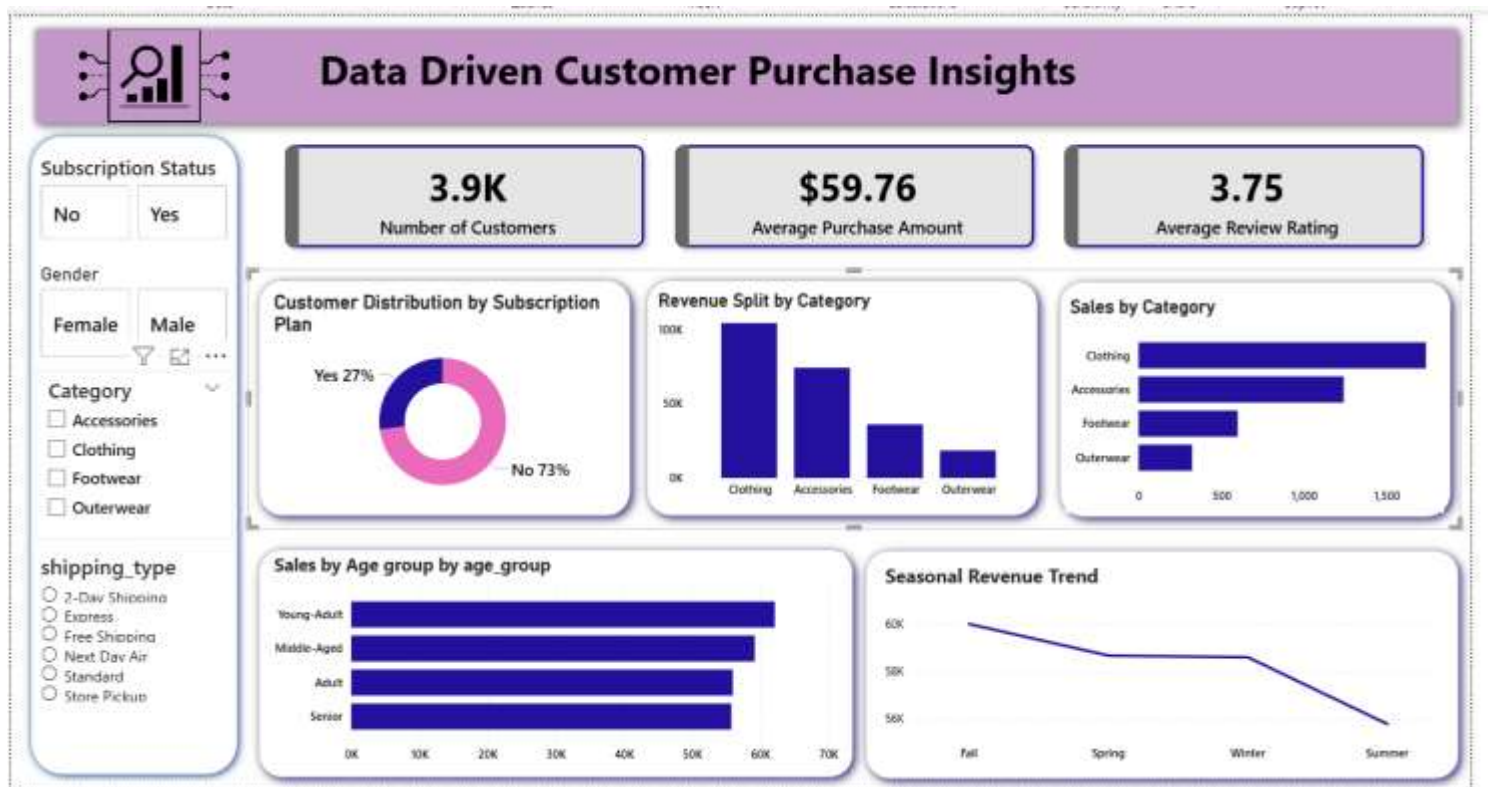


Data Pipeline:

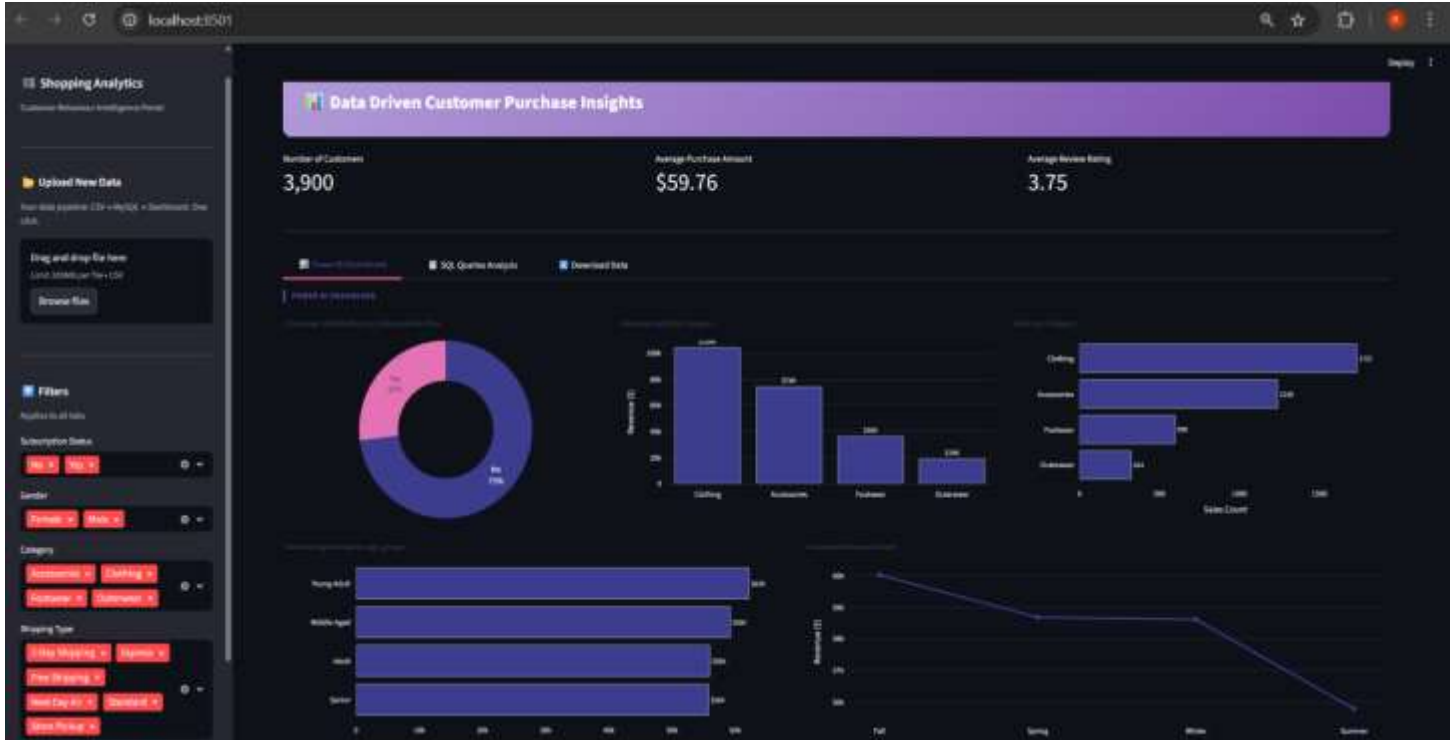
Analytics Pipeline



PowerBi Dashboard:



Analytical Web Portal:



Description :

- **Tools Used:** Streamlit, Python, Pandas, Plotly, MySQL
- **Database Connection:** Connects to MySQL to fetch and store data
- **Data Cleaning:** Handles missing values and formats data
- **Feature Engineering:** Creates new features like age_group and total_spend
- **Data Visualization:** Displays charts such as bar, pie, and line graphs
- **Dashboard Features:** Includes KPI metrics, filters, and multiple tabs
- **SQL Analysis:** Performs queries to generate insights and recommendations
- **Data Upload & Download:** Allows users to upload CSV and download data

Bibliography:

Dataset: <https://www.kaggle.com/>

<https://www.geeksforgeeks.org/pandas/pandas-tutorial/>

Plotly Python Documentation: <https://plotly.com/python>

Streamlit Documentation: <https://docs.streamlit.io>